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Abstract

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I. MAJORANA EXPANSION OF A TWO-PARTICLE OPERATOR

Majorana Operator Basis

1. The Majorana operators γ_i are Hermitian and satisfy the Clifford algebra:

$$\{x_j, x_k\} = 2\delta_{jk}I_{\mathcal{H}_{\mathcal{F}}}, \quad 1 \leq j, k \leq 2r$$

hence

$$x_j^2 = I_{\mathcal{H}_{\mathcal{F}}}, \quad 1 \leq j \leq 2r \quad (1)$$

and

$$x_j x_k = -x_k x_j, \quad 1 \leq j, k \leq 2r \quad (2)$$

For a system with r fermionic modes, $2r$ Majorana operators are defined in terms of creation $\{a_j^\dagger, 1 \leq j \leq 2r\}$ and annihilation $\{a_j, 1 \leq j \leq 2r\}$ operators:

$$x_{2j-1} = a_j + a_j^\dagger, \quad x_{2j} = -i(a_j - a_j^\dagger), \quad 1 \leq j \leq 2r$$

giving

$$(\mathbf{x}) = \begin{pmatrix} x_1 & x_2 \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \mathbb{I}_r & i\mathbb{I}_r \\ \mathbb{I}_r & -i\mathbb{I}_r \end{pmatrix} \quad (3)$$

and

$$\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{I}_r \\ i\mathbb{I}_r & -i\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger \\ \mathbf{a} \end{pmatrix}. \quad (4)$$

2. The inverse relations are:

$$a_i = \frac{1}{2}(x_{2i-1} + ix_{2i}), \quad a_j^\dagger = \frac{1}{2}(x_{2i-1} - ix_{2i})$$

giving

$$(\mathbf{a}^\dagger, \mathbf{a}) = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} \mathbb{I}_r & \mathbb{I}_r \\ -i\mathbb{I}_r & i\mathbb{I}_r \end{pmatrix} \quad (5)$$

and

$$\begin{pmatrix} \mathbf{a}^\dagger \\ \mathbf{a} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbb{I}_r & -i\mathbb{I}_r \\ \mathbb{I}_r & i\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix}. \quad (6)$$

3. In particular

$$\begin{pmatrix} \mathbf{a}^\dagger & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} \mathbb{I}_r & \mathbb{O} \\ -i\mathbb{I}_r & \mathbb{O} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbf{x}_1 - i\mathbf{x}_2 & \mathbf{0} \end{pmatrix} \quad (7)$$

$$\begin{pmatrix} \mathbf{a} \\ \mathbf{0} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbb{I}_r & i\mathbb{I}_r \\ \mathbb{O} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbf{x}_1 + i\mathbf{x}_2 \\ \mathbf{0} \end{pmatrix}. \quad (8)$$

4. This gives one particle particle conserving operators in super vector form as

$$(\mathbf{a}^\dagger \mathbf{a}) = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{0} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbf{x}_1 - i\mathbf{x}_2 & \mathbf{0} \end{pmatrix} \frac{1}{2} \begin{pmatrix} \mathbf{x}_1 + i\mathbf{x}_2 \\ \mathbf{0} \end{pmatrix} = \frac{1}{4} (\mathbf{x}_1 - i\mathbf{x}_2) (\mathbf{x}_1 + i\mathbf{x}_2) \quad (9)$$

$$= \frac{1}{4} (\mathbf{x}_1 \mathbf{x}_1 + i\mathbf{x}_1 \mathbf{x}_2 - i\mathbf{x}_2 \mathbf{x}_1 + \mathbf{x}_2 \mathbf{x}_2) = \frac{1}{4} (\mathbf{x}_1 \mathbf{x}_1 + \mathbf{x}_2 \mathbf{x}_2 + i(\mathbf{x}_1 \mathbf{x}_2 - \mathbf{x}_2 \mathbf{x}_1)) \quad (10)$$

giving the following expression for individual operators

$$a_j^\dagger a_k = \frac{1}{4} (\mathbf{x}_j \mathbf{x}_k + i\mathbf{x}_j \mathbf{x}_{k+r} - i\mathbf{x}_{j+r} \mathbf{x}_k + \mathbf{x}_{j+r} \mathbf{x}_{k+r}) = \frac{1}{4} (\mathbf{x}_j \mathbf{x}_k + \mathbf{x}_{j+r} \mathbf{x}_{k+r} + i(\mathbf{x}_j \mathbf{x}_{k+r} - \mathbf{x}_{j+r} \mathbf{x}_k)), \quad (11)$$

5. This gives two particle conserving operators in super vector form as

$$(\mathbf{a}^\dagger \mathbf{a}^\dagger \mathbf{a} \mathbf{a}) = (\mathbf{a}^\dagger) (\mathbf{a}^\dagger) (\mathbf{a}) (\mathbf{a}) = \left(\begin{pmatrix} a_1^\dagger & \cdots & a_r^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & \cdots & a_r^\dagger \end{pmatrix} \right) \left(\begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} \right) \quad (12)$$

$$= \frac{1}{16} \left(\begin{pmatrix} \mathbf{x}_1 - i\mathbf{x}_2 & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 - i\mathbf{x}_2 & \mathbf{0} \end{pmatrix} \right) \left(\begin{pmatrix} \mathbf{x}_3 + i\mathbf{x}_4 \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x}_3 + i\mathbf{x}_4 \\ \mathbf{0} \end{pmatrix} \right) \quad (13)$$

$$= \frac{1}{16} \left(\begin{pmatrix} (\mathbf{x}_1 - i\mathbf{x}_2) (\mathbf{x}_1 - i\mathbf{x}_2) & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \right) \begin{pmatrix} (\mathbf{x}_3 + i\mathbf{x}_4) (\mathbf{x}_3 + i\mathbf{x}_4) \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix} \quad (14)$$

$$= \frac{1}{16} ((\mathbf{x}_1 - i\mathbf{x}_2) (\mathbf{x}_1 - i\mathbf{x}_2) (\mathbf{x}_3 + i\mathbf{x}_4) (\mathbf{x}_3 + i\mathbf{x}_4)) \quad (15)$$

giving the following expression for individual operators

$$a_j^\dagger a_k^\dagger a_l a_m = \frac{1}{16} (x_j - ix_{2j})(x_{2k-1} - ix_{2k})(x_{2l-1} + ix_{2l})(x_{2m-1} + ix_{2m}),$$

$$1 \leq j < k \leq r, 1 \leq l < m \leq r$$

that number

$$\binom{r}{2}^2 = \frac{r^2}{4} (r-1)^2 \quad (16)$$

In particular these operators can be classified into:

(a) Operators that can be described by two unequal subscripts

$$a_j^\dagger a_k^\dagger a_k a_j = n_j n_k = \frac{1}{2} (I_{\mathcal{H}_{\mathcal{F}}} + ix_j x_{j+r}) \frac{1}{2} (I_{\mathcal{H}_{\mathcal{F}}} + ix_k x_{k+r})$$

$$\parallel$$

$$\frac{1}{4} (I_{\mathcal{H}_{\mathcal{F}}} + i(x_j x_{j+r} + x_k x_{k+r}) - x_j x_{j+r} x_k x_{k+r}),$$

that number

$$\binom{r}{2} = \frac{r}{2} (r-1) = \frac{r(r-1)}{(1)(2)}. \quad (17)$$

In the case $r = 4$ the subscripts are

$$1212, 1313, 1414, 2323, 2424, 3434 \quad (18)$$

(b) Operators that can be described by three unequal subscripts

$$a_j^\dagger a_l a_k^\dagger a_k = a_j^\dagger a_l n_k = n_k a_j^\dagger a_l \quad (19)$$

$$\parallel \quad (20)$$

$$\frac{1}{8} \left\{ x_j x_l + i x_{j+r} x_{l+r} - i x_{j+r} x_l + x_{j+r} x_{l+r} + i x_j x_l x_k x_{k+r} - x_j x_{l+r} x_k x_{k+r} + x_{j+r} x_l x_k x_{k+r} + i x_j x_l x_k x_{k+r} \right. \quad (21)$$

$$\parallel \quad (22)$$

$$\frac{1}{8} \left\{ x_j x_l + x_{j+r} x_{l+r} - x_j x_{l+r} x_k x_{k+r} + x_{j+r} x_l x_k x_{k+r} + i (x_{j+r} x_{l+r} - x_{j+r} x_l) + i (x_j x_l x_k x_{k+r} + \right. \quad (23)$$

that number

$$\# = 2 \binom{r-1}{2} \binom{r}{1} = (r-1)(r-2)(r) = (r-1)(r-2)(r) = r(r-1)(r-2). \quad (24)$$

In the case $r = 4$ the subscripts are

$$1233, 1433, 2433, 2133, 4133, 4233, 3312, 3314, 3324, 3321, 3341, 3342 \quad (25)$$

Remark 1 Operators that can be described by four unequal subscripts

$$a_i^\dagger a_j^\dagger a_k a_l = a_i^\dagger a_l a_j^\dagger a_k \quad (26)$$

$$\parallel \quad (27)$$

$$\frac{1}{16} \left\{ (x_i x_j x_k x_l + x_i x_j x_{k+r} x_{l+r} - x_i x_{j+r} x_k x_{l+r} + x_i x_{j+r} x_{k+r} x_l + x_{i+r} x_j x_k x_{l+r} - x_{i+r} x_j x_{k+r} x_l + x_{i+r} x_{j+r} x_k x_{l+r} \right. \quad (28)$$

$$+ i (x_i x_j x_k x_{l+r} - x_i x_j x_{k+r} x_l + x_i x_{j+r} x_k x_l + x_i x_{j+r} x_{k+r} x_{l+r} + x_i x_{j+r} x_k x_l + x_{i+r} x_j x_{k+r} x_{l+r} + x_{i+r} x_{j+r} x_k x_{l+r} \quad (29)$$

that number

$$\# = \binom{2r}{2} \binom{2r-2}{2} = \frac{r}{2} (2r-1)(2r-2)(2r-3). \quad (30)$$

In the case $r = 2$ the subscripts of the Majorana operators are

$$1234, 1324, 1423, 2314, 2413, 3412.$$

$$\sigma(\{1, 2, 3, 4\}), \quad \sigma \in C_4^{2r} \subset S_{2r}, \quad \sigma = \alpha(\{j, k\}) \beta(\{l, m\}), \quad \alpha \in C_2^{2r} \subset S_r, \quad \beta \in C_2^{2r-2} \subset S_{r-2} \quad (31)$$

$$S_2 \times S_{2r-2} \subset S_{2r} \quad (32)$$

Definition 2 *The complex vector space of two particle operators*

$$\mathcal{A}_{a2} = \mathbb{C}\text{-linspan} \left\{ a_j^\dagger a_k^\dagger a_l a_m \mid 1 \leq j < k \leq r, 1 \leq m < l \leq r \right\}. \quad (33)$$

Definition 3 *Decomposition of $\mathcal{A}_{a2}$*

1. *Two out of four subscripts different*

(a) *The subspace of operators $\mathcal{A}_{a2(2)} \subset \mathcal{A}_{a2}$*

$$\mathcal{A}_{a2(2)} = \mathbb{C}\text{-linspan} \left\{ a_j^\dagger a_k^\dagger a_k a_j \mid 1 \leq j < k \leq r \right\}. \quad (34)$$

(b) *The index set $\mathfrak{J}_2^4$*

$$\mathfrak{J}_2^4 = \{ (j, k) \mid 1 \leq j < k \leq r \}. \quad (35)$$

2. *Three subscripts out of four different*

(a) *The subspace of operators $\mathcal{A}_{a2(3)} \subset \mathcal{A}_{a2}$*

$$\mathcal{A}_{a2(3)} = \mathbb{C}\text{-linspan} \left\{ a_j^\dagger a_k^\dagger a_l a_m \mid 1 \leq k \neq j \neq l \leq r \right\} \quad (36)$$

(b) *The index set $\mathfrak{J}_3^4$*

$$\mathfrak{J}_3^4 = \{ (j, l, k) \mid 1 \leq k \neq j \neq l \leq r \} \quad (37)$$

3. *Four subscripts out of four different*

(a) *The subspace of operators $\mathcal{A}_{a2(4)} \subset \mathcal{A}_{a2}$*

$$\mathcal{A}_{a2(4)} = \mathbb{C}\text{-linspan} \left\{ a_j^\dagger a_k^\dagger a_l a_m \mid 1 \leq j < k \leq r, 1 \leq l < m \leq r, (j, k) \neq (l, m) \right\} \quad (38)$$

(b) *The index set $\mathfrak{J}_4^4$*

$$\mathfrak{J}_4^4 = \{ (j, k, l, m) \mid 1 \leq j < k \leq r, 1 \leq l < m \leq r, (j, k) \neq (l, m) \} \quad (39)$$

Lemma 4 *The three subspaces $\{\mathcal{A}_{a2(2)}, \mathcal{A}_{a2(3)}, \mathcal{A}_{a2(4)}\}$ are disjoint and*

$$\mathcal{A}_{a2} = \mathcal{A}_{a2(2)} \oplus \mathcal{A}_{a2(3)} \oplus \mathcal{A}_{a2(4)}. \quad (40)$$

Lemma 5 *The vector spaces $\{\mathcal{A}_{a2}, \mathcal{A}_{a2(2)}, \mathcal{A}_{a2(3)}, \mathcal{A}_{a2(4)}\}$ have the dimensions*

1.

$$\dim \{\mathcal{A}_{a2}\} = \binom{r}{2}^2 \quad (41)$$

2.

$$\dim \{\mathcal{A}_{a2(2)}\} = \binom{r}{2} \quad (42)$$

3.

$$\dim \{\mathcal{A}_{a2(3)}\} = 2 \binom{r-1}{2} \binom{r}{1} \quad (43)$$

4.

$$\dim \{\mathcal{A}_{a2(4)}\} = \binom{r}{2} \binom{r-2}{2}. \quad (44)$$

Lemma 6

$$\binom{r}{2}^2 = \binom{r}{2} + 2 \binom{r-1}{2} \binom{r}{1} + \binom{r}{2} \binom{r-2}{2} \quad (45)$$

Proof.

1. (a)

$$\binom{r}{2} = \frac{r(r-1)}{2} = \frac{1}{2}r^2 - \frac{1}{2}r = \frac{1}{4}(2r^2 - 2r) \quad (46)$$

(b) In the case $r = 4$

$$\frac{4(4-1)}{2} = 6. \quad (47)$$

(c) In general

$$\binom{r}{2}^2 = \frac{r(r-1)}{2} \frac{r(r-1)}{2} = \frac{1}{4}r^2(r^2 - 2r + 1) = \frac{1}{4}r^4 - \frac{1}{2}r^3 + \frac{1}{4}r^2 = \frac{1}{4}(r^4 - 2r^3 + r^2) \quad (48)$$

b In the case $r = 4$

$$\frac{1}{4}4^4 - \frac{1}{2}4^3 + \frac{1}{4}4^2 = 64 - 32 + 4 = 36 \quad (49)$$

(d) In general

$$\binom{r}{2} = \frac{r(r-1)}{2} = \frac{1}{2}r^2 - \frac{1}{2}r = \frac{1}{4}(2r^2 - 2r) \quad (50)$$

(e) In the case $r = 4$

$$\frac{4(4-1)}{2} = 6 \quad (51)$$

(f) In general

$$2 \binom{r-1}{2} \binom{r}{1} = r(r-1)(r-2) = r(r^2 - 3r + 2) = r^3 - 3r^2 + 2r = \frac{1}{4}(4r^3 - 12r^2 + 8r) \quad (52)$$

(g) In the case $r = 4$

$$4^3 - (3)4^2 + (2)4 = 64 - 48 + 8 = 24 \quad (53)$$

$$2 \binom{4-1}{2} \binom{4}{1} = 2 \binom{3}{2} \binom{4}{1} = 4(4-1)(4-2) = 4(4^2 - (3)4 + 2) = 4^3 - 3(4^2) + 2(4) \quad (54)$$

(h) In general

$$\binom{r}{2} \binom{r-2}{2} = \frac{1}{4} \left(r^2 - \frac{1}{2}r \right) (r-2)(r-3) = \frac{1}{4} (r^2 - r)(r^2 - 5r + 6) \quad (55)$$

$$= \frac{1}{4} (r^2(r^2 - 5r + 6) - r(r^2 - 5r + 6)) = \frac{1}{4} (r^4 - 5r^3 + 6r^2 - r^3 + 5r^2 - 6r) \quad (56)$$

$$= \frac{1}{4} (r^4 - 6r^3 + 11r^2 - 6r) \quad (57)$$

(i) In the case $r = 4$

$$\frac{1}{4} 4(4^3 - 6(4^2) + 11(4) - 6) = 64 - 96 + 44 - 6 = 108 - 102 = 6 \quad (58)$$

2. The sum of these three terms are

$$\frac{1}{4} (2r^2 - 2r) + \frac{1}{4} (4r^3 - 12r^2 + 8r) + \frac{1}{4} (r^4 - 6r^3 + 11r^2 - 6r) \quad (59)$$

$$= \frac{1}{4} (2r^2 - 2r + 4r^3 - 12r^2 + 8r + r^4 - 6r^3 + 11r^2 - 6r) \quad (60)$$

$$= \frac{1}{4} (r^2 - 2r^3 + r^4), \quad (61)$$

which equals $\binom{r}{2}^2$ i.e.

$$\frac{1}{4} (r^4 - 2r^3 + r^2). \blacksquare \quad (62)$$

■

1. **Example 7** Let $r = 4$ then

(a) The full basis has cardenality given by

i.

$$\binom{r}{2}^2 = \frac{r^2}{4} (r-1)^2 = \frac{16}{4} (4-1)^2 = (4)(9) = 36 \quad (63)$$

in detail

ii.

$$a_j^\dagger a_k^\dagger a_l a_m \quad (64)$$

$$\parallel \quad (65)$$

$$1212, 1213, 1214, 1223, 1224, 1234, \quad 1312, 1313, 1314, 1323, 1324, 1334, \quad 1412, 1413, 1414, \quad (66)$$

$$2312, 2313, 2314, 2323, 2324, 2334, \quad 2412, 2413, 2414, 2423, 2424, 2434, \quad 3412, 3413, 3414, \quad (67)$$

(b) The subset of the basis describes by two subscripts has cardenality given by

i.

$$\binom{r}{2} = \frac{r}{2} (r-1) = \left(\frac{4(4-1)}{(1)(2)} \right) = 6 \quad (68)$$

in detail

ii.

$$a_j^\dagger a_j a_k^\dagger a_k = n_j n_k \quad (69)$$

$$\parallel \quad (70)$$

$$1212, \quad 1313, \quad 1414, \quad 2323, \quad 2424, \quad 3434 \quad (71)$$

(c) The subset of the basis describes by three subscripts has cardenality given by

i.

$$2 \binom{r-1}{2} \binom{r}{1} = 2 \frac{(r-1)(r-2)}{2} (r) = (r-1)(r-2)(r) = (4-1)(4-2)(4) = (3)(2) \quad (72)$$

in detail

ii.

$$a_j^\dagger a_k a_l^\dagger a_l = a_j^\dagger a_k n_l \quad (73)$$

$$\parallel \quad (74)$$

$$1233, 1433, 2433, 2133, 4133, 4233, \quad 1244, 1344, 2344, 2144, 3144, 3244, \quad (75)$$

$$1322, 1422, 3422, 3122, 4122, 4322, \quad 4211, 4311, 2311, 2411, 3411, 3211, \quad (76)$$

(d) The subset of the basis describes by four subscripts has cardenality given by

i.

$$\binom{r}{2} \binom{r-2}{2} = \binom{4}{2} \binom{2}{2} = (6)(1) = 6 \quad (77)$$

in detail

ii.

$$a_j^\dagger a_k^\dagger a_l a_m = -a_j^\dagger a_l a_k^\dagger a_m \quad (78)$$

$$\parallel \quad (79)$$

$$1234, 1324, 1423, 2314, 2413, 3412. \quad (80)$$

2. General Two-Particle Operator in Second Quantization

A general two-particle operator $\hat{O}$ is written as:

$$\hat{O} = \sum_{i,j,k,l} O_{ijkl} a_j^\dagger a_j^\dagger a_k a_l = \frac{1}{4} \sum_{i < j < k < l} O_{ijkl} a_j^\dagger a_j^\dagger a_k a_l,$$

where O_{ijkl} are antisymmetric under $i \leftrightarrow j$ or $k \leftrightarrow l$. In the special case $a_{i_0}^\dagger a_{j_0}^\dagger a_{k_0} a_{l_0}$ the coefficients are

$$O_{i_0 j_0 k_0 l_0} = \delta_{i i_0} \delta_{j j_0} \delta_{k k_0} \delta_{l l_0} \quad (81)$$

Example 8 Coulomb Interaction

For electrons in a basis $\{\phi_j(\mathbf{r})\}$:

$$O_{jklm} = V_{jklm} = \int d\mathbf{r}_1 d\mathbf{r}_2 \phi_j^*(\mathbf{r}_1) \phi_k^*(\mathbf{r}_2) \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \phi_l(\mathbf{r}_1) \phi_m(\mathbf{r}_2).$$

Basis Construction

The field operator basis for second degree particle conserving operators is

$$(\mathbf{a}^\dagger \mathbf{a}^\dagger \mathbf{a} \mathbf{a}) = \left(a_1^\dagger a_2^\dagger a_1 a_2 \cdots a_{r-1}^\dagger a_r^\dagger a_{r-1} a_r \right) \quad (82)$$

such that

$$\text{cardenality} = \left(\frac{r}{2} (r-1) \ r (r-1) (r-2) \ \frac{r}{4} (r-1) (r-2) (r-3) \right), \quad (83)$$

which can be expressed in Majorana form as

$$\left\{ \left\{ \frac{1}{4} (I_{\mathcal{H}_{\mathcal{F}}} + i(x_j x_{j+r} + x_k x_{k+r}) - x_j x_{j+r} x_k x_{k+r}) \right\} \middle| 1 \leq j < k \leq r \right\}, \quad (84)$$

$$\left\{ \frac{1}{8} \left(x_j x_l + x_{j+r} x_{l+r} - x_j x_{l+r} x_k x_{k+r} + x_{j+r} x_l x_k x_{k+r} + i(x_{j+r} x_{l+r} - x_{j+r} x_l) + i(x_j x_l x_k x_{k+r} + x_{j+r} x_{l+r} x_k x_{k+r}) \right) \right. \\ \left. j, k, l, \right. \quad (85)$$

$$\frac{1}{16} \left\{ (x_i x_j x_k x_l + x_i x_j x_{k+r} x_{l+r} - x_i x_{j+r} x_k x_{l+r} + x_i x_{j+r} x_{k+r} x_l + x_{i+r} x_j x_k x_{l+r} - x_{i+r} x_j x_{k+r} x_l + x_{i+r} x_{j+r} x_k x_{l+r} \right. \quad (86)$$

$$+ i(x_i x_j x_k x_{l+r} - x_i x_j x_{k+r} x_l + x_i x_{j+r} x_k x_l + x_i x_{j+r} x_{k+r} x_{l+r} + x_i x_{j+r} x_k x_l + x_{i+r} x_j x_{k+r} x_{l+r} + x_{i+r} x_{j+r} x_k x_{l+r}) \\ \left. i, j, k, l, \right. \quad (87)$$

such that

$$\text{cardenality} = \left(\frac{r}{2} (r-1) \frac{r}{2} (r-1) (r-2) \frac{r}{4} (r-1) (r-2) (r-3) \right). \quad (88)$$

Remark 9

$$\hat{O} = \hat{O}_{(2)} + \hat{O}_{(3)} + \hat{O}_{(4)} \quad (89)$$

Remark 10

$$\hat{O} = \frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} \left(I_{\mathcal{H}_{\mathcal{F}}} + i(x_j x_{j+r} + x_k x_{k+r}) - x_j x_{j+r} x_k x_{k+r} \right) \quad (90)$$

$$+ \frac{1}{8} \sum_{j \neq k \neq l} O_{jkjl} \left(x_j x_l + x_{j+r} x_{l+r} + i(x_j x_{l+r} - x_{j+r} x_l) \right) \quad (91)$$

$$+ x_{j+r} x_l x_k x_{k+r} - x_j x_{l+r} x_k x_{k+r} + i(x_{j+r} x_{l+r} x_k x_{k+r} + x_j x_l x_k x_{k+r}) \quad (92)$$

$$+ \frac{1}{16} \sum_{i \neq j \neq k \neq l} O_{ijkl} \left(\begin{aligned} & x_i x_j x_k x_l + x_i x_j x_{k+r} x_{l+r} - x_i x_{j+r} x_k x_{l+r} + x_i x_{j+r} x_{k+r} x_l \\ & + x_{i+r} x_j x_k x_{l+r} - x_{i+r} x_j x_{k+r} x_l + x_{i+r} x_{j+r} x_k x_l + x_{i+r} x_{j+r} x_{k+r} x_{l+r} \end{aligned} \right) \quad (93)$$

$$+ i \left(\begin{aligned} & x_i x_{j+r} x_k x_l + x_i x_{j+r} x_{k+r} x_{l+r} - x_{i+r} x_j x_k x_l + x_i x_j x_k x_{l+r} \\ & - x_i x_j x_{k+r} x_l + x_{i+r} x_j x_{k+r} x_{l+r} + x_{i+r} x_{j+r} x_k x_{l+r} - x_{i+r} x_{j+r} x_{k+r} x_l \end{aligned} \right) \quad (94)$$

where

1.

$$\hat{O}_{(2)} = \frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} \left(I_{\mathcal{H}_{\mathcal{F}}} + i(x_j x_{j+r} + x_k x_{k+r}) - x_j x_{j+r} x_k x_{k+r} \right). \quad (95)$$

2.

$$\hat{O}_{(3)} = \frac{1}{8} \sum_{j \neq k \neq l} O_{jkjl} \left(x_j x_l + x_{j+r} x_{l+r} + i(x_j x_{l+r} - x_{j+r} x_l) \right) \quad (96)$$

$$+ x_{j+r} x_l x_k x_{k+r} - x_j x_{l+r} x_k x_{k+r} + i(x_{j+r} x_{l+r} x_k x_{k+r} + x_j x_l x_k x_{k+r}) \quad (97)$$

3. change

$$\hat{O}_{(4)} = \frac{1}{16} \sum_{j \neq k \neq l \neq m} O_{jklm} \left(\begin{aligned} & x_j x_k x_l x_m + x_j x_k x_{l+r} x_{m+r} - x_j x_{k+r} x_l x_{m+r} + x_j x_{k+r} x_{l+r} x_m \\ & + x_{j+r} x_k x_l x_{m+r} - x_{j+r} x_k x_{l+r} x_m + x_{j+r} x_{k+r} x_l x_m + x_{j+r} x_{k+r} x_{l+r} x_{m+r} \end{aligned} \right) \quad (98)$$

$$+ j \left(\begin{aligned} & x_j x_{k+r} x_l x_m + x_j x_{k+r} x_{l+r} x_{m+r} - x_{j+r} x_k x_l x_m + x_j x_k x_l x_{m+r} \\ & - x_j x_k x_{l+r} x_m + x_{j+r} x_k x_{l+r} x_{m+r} + x_{j+r} x_{k+r} x_l x_{m+r} - x_{j+r} x_{k+r} x_{l+r} x_m \end{aligned} \right), \quad (99)$$

Remark 11 The terms of similar degree can be collected together to give

$$\hat{O} \quad (100)$$

$$\parallel \quad (101)$$

$$\frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} I_{\mathcal{H}_{\mathcal{F}}} \quad (102)$$

$$+ \quad (103)$$

$$\frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} i \left(i(x_j x_{j+r} + x_k x_{k+r}) \right) + \frac{1}{8} \sum_{j \neq k \neq l} O_{jkjl} \left(x_j x_l + x_{j+r} x_{l+r} + i(x_j x_{l+r} - x_{j+r} x_l) \right) \quad (104)$$

$$+ \quad (105)$$

$$- \frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} x_j x_{j+r} x_k x_{k+r} + \frac{1}{8} \sum_{j \neq k \neq l} O_{jkjl} \left(x_{j+r} x_l x_k x_{k+r} - x_j x_{l+r} x_k x_{k+r} + i(x_j x_l x_k x_{k+r} + x_{j+r} x_{l+r} x_k x_{k+r}) \right) \quad (106)$$

$$+ \quad (107)$$

$$\frac{1}{16} \sum_{i \neq j \neq k \neq l} O_{ijkl} \left(\begin{aligned} & x_i x_j x_k x_l + x_i x_j x_{k+r} x_{l+r} - x_i x_{j+r} x_k x_{l+r} + x_i x_{j+r} x_{k+r} x_l \\ & + x_{i+r} x_j x_k x_{l+r} - x_{i+r} x_j x_{k+r} x_l + x_{i+r} x_{j+r} x_k x_l + x_{i+r} x_{j+r} x_{k+r} x_{l+r} \end{aligned} \right) \quad (108)$$

$$+ \left(\begin{aligned} & i x_i x_{j+r} x_{k+r} x_{l+r} - i x_{i+r} x_j x_k x_l + i x_i x_j x_k x_{l+r} - i x_i x_j x_{k+r} x_l \\ & + i (x_i x_{j+r} x_k x_l + x_{i+r} x_j x_{k+r} x_{l+r} + x_{i+r} x_{j+r} x_k x_{l+r} - x_{i+r} x_{j+r} x_{k+r} x_l) \end{aligned} \right) \quad (109)$$

1.

$$\frac{1}{4} \sum_{1 \leq j < k \leq r} O_{jkjk} I_{\mathcal{H}_{\mathcal{F}}} = \frac{1}{8} \text{Tr} \left(\hat{O} \right) \quad (110)$$

2.

$$\frac{i}{4} \sum_{(j,k) \in \mathcal{I}_3^4} O_{jkjk} (x_j x_{j+r} + x_k x_{k+r}) = \frac{i}{2} \sum_{(j,k) \in \mathcal{I}_3^4} O_{jkjk} x_j x_{j+r} \quad (111)$$

3.

$$\frac{1}{8} \sum_{(j,k,l) \in \mathcal{I}_3^4} \left\{ O_{jkjl} x_j x_l + O_{jkjl} x_{j+r} x_{l+r} + i (O_{jkjl} x_j x_{l+r} - O_{jkjl} x_{j+r} x_l) \right\} \quad (112)$$

$$= \frac{1}{8} \sum_{(j,k,l) \in \mathcal{I}_3^4} \left\{ O_{jkjl} x_j x_l + O_{jkjl} x_{j+r} x_{l+r} + i (O_{jkjl} x_j x_{l+r} - O_{jkjl} x_{j+r} x_l) \right\} \quad (113)$$

4.

$$\frac{1}{16} \sum_{(i,j,k,l) \in \mathcal{I}_4^4} O_{ijkl} \left\{ (x_i x_j x_k x_l + x_i x_j x_{k+r} x_{l+r} - x_i x_{j+r} x_k x_{l+r} + x_i x_{j+r} x_{k+r} x_l \right. \quad (114)$$

$$\left. + x_{i+r} x_j x_k x_{l+r} - x_{i+r} x_j x_{k+r} x_l + x_{i+r} x_{j+r} x_k x_l + x_{i+r} x_{j+r} x_{k+r} x_{l+r} \right) \quad (115)$$

$$+ i (x_i x_j x_k x_{l+r} - x_i x_j x_{k+r} x_l + x_i x_{j+r} x_k x_l + x_i x_{j+r} x_{k+r} x_{l+r} \quad (116)$$

$$\left. + x_i x_{j+r} x_k x_l + x_{i+r} x_j x_{k+r} x_{l+r} + x_{i+r} x_{j+r} x_k x_{l+r} - x_{i+r} x_{j+r} x_{k+r} x_l) \right\} \quad (117)$$

3. Expansion in Majorana Operators

Substituting $a_j, a_j^\dagger$ in terms of x_i , we expand $\hat{O}$ into products of up to four Majorana operators, grouped by degree:

Degree 0 (Constant Term)

$$CI_{\mathcal{H}_{\mathcal{F}}} = \text{Tr}(\hat{O})I_{\mathcal{H}_{\mathcal{F}}} = \frac{1}{2^r} \text{Tr}(\hat{O})I_{\mathcal{H}_{\mathcal{F}}} = \frac{1}{8} \sum_{i,j} O_{ijji} I_{\mathcal{H}_{\mathcal{F}}} \quad (\text{as previously derived}).$$

Degree 2 (Quadratic Terms)

After full antisymmetrization and index relabeling:

$$\hat{O}_2 = \sum_{j < k} \alpha_{jk} x_j x_k, \quad (118)$$

where

$$\alpha_{jk} = \frac{1}{4} \sum_{r,s} \begin{cases} \text{Re}(O_{jrsk} - O_{jsrk}) & \text{if } j = 2m - 1, k = 2n - 1 \\ \text{Re}(O_{jrsk} + O_{jsrk}) & \text{if } j = 2m, k = 2n \\ -i \text{Im}(O_{jrsk} - O_{jsrk}) & \text{if } j = 2m - 1, k = 2n \\ +i \text{Im}(O_{jrsk} + O_{jsrk}) & \text{if } j = 2m, k = 2n - 1 \end{cases}.$$

Degree 4 (Quartic Terms)

Fully uncontracted products of four Majorana operators:

$$\hat{O}_4 = \sum_{i < j < k < l} \beta_{ijkl} x_i x_j x_k x_l,$$

where the coefficients β_{ijkl} are given by the fully antisymmetrized expression becomes:

$$\beta_{ijkl} = \frac{1}{16} \sum_{\substack{\sigma \in S_4 \\ \text{distinct } i,j,k,l}} \text{sgn}(\sigma) \tilde{O}_{\sigma(p)\sigma(q)\sigma(r)\sigma(s)},$$

where $\tilde{O}_{pqrs}$ maps fermionic indices to Majorana indices:

$$\tilde{O}_{pqrs} = \begin{cases} O_{pqrs} & \text{even } x\text{-parity} \\ -iO_{pqrs} & \text{odd } x\text{-parity} \end{cases}$$

The quartic coefficient β_{ijkl} in the Majorana expansion is given by the fully antisymmetrized combination:

$$\beta_{ijkl} = \frac{1}{16} \sum_{\sigma \in S_4} \text{sgn}(\sigma) \tilde{O}_{\sigma(p)\sigma(q)\sigma(r)\sigma(s)}$$

where:

- S_4 is the symmetric group of all permutations of 4 elements
- $\text{sgn}(\sigma)$ is the sign of the permutation (+1 for even, -1 for odd)
- The indices (i, j, k, l) correspond to the Majorana operator ordering
- (p, q, r, s) are the original fermionic mode indices mapped to $(2p-1, 2p, 2q-1, 2q)$.

This represents the sum over all 24 possible permutations of the four indices with appropriate sign factors.

4. Final Majorana Expansion

Remark 12 *The expansion of two particle operators in terms of the corresponding Majorana operators can be organized into homogeneous terms of zeroth, second and fourth degree, but these terms are **not** linearly independent.*

Combining all terms, the operator $\hat{O}$ becomes:

$$\hat{O} = \sum_{i < j} \alpha_{ij} x_i x_j + \sum_{i < j < k < l} \beta_{ijkl} x_i x_j x_k x_l.$$

5. Key Observations

- Quadratic terms (α_{ij}) depend on off-diagonal elements of O_{ijkl} .
 - Quartic terms (β_{ijkl}) are fully antisymmetrized four-Majorana interactions.
- Consistency Checks

1. **Antisymmetry Preservation:** All coefficients satisfy:

$$\alpha_{pq} = -\alpha_{qp}, \quad \beta_{ijkl} = \text{totally antisymmetric.}$$

2. Trace Conditions:

$$\text{Tr}(x_i x_j) = 0, \quad \text{Tr}(x_i x_j x_k x_l) = 0 \quad \Rightarrow \quad \text{Tr}(\hat{O}) = 2^N C.$$

3. Example: Hubbard Interaction: For $U \sum_i n_{i\uparrow} n_{i\downarrow}$:

$$C = \frac{UN}{8}, \quad \alpha_{2i-1,2i} = \frac{U}{4}, \quad \text{other } \alpha, \beta = 0.$$